

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL-2 Industry Problem Solving

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 –INDUSTRY PROBLEM SOLVING	Assessment:	Assessment Tool 2–INDUSTRY PROBLEM SOLVING
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
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Section / Batch:		Date:	18/08/2026

Code of Conduct

I _____POOJA S(192524211)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____POOJA S_____

1. A cybersecurity system historically detects 95% of malicious activities. After deploying a new detection algorithm, it correctly detects 194 out of 200 attacks. Test whether the new algorithm has significantly improved the detection rate using a p-value.

Answer:

a. Identify the Population and Sample

Population: All malicious activities or cyberattacks that can be detected by the new cybersecurity system.

Sample: 200 attacks tested using the new detection algorithm. Out of these, 194 attacks were correctly detected.

b. Identify the Parameter of Interest

The parameter of interest is the true proportion of malicious activities detected by the new algorithm.

Let:

p = True detection rate of the new cybersecurity algorithm.

c. Calculate the Point Estimate

Sample proportion,

$$\hat{p} = x / n$$

$$\hat{p} = 194 / 200$$

$$\hat{p} = 0.97$$

Therefore, the point estimate of the detection rate is:

$$\hat{p} = 97\%$$

d. State the Null Hypothesis (H_0)

$$H_0: p = 0.95$$

The null hypothesis states that the new algorithm has not significantly improved the detection rate.

e. State the Alternative Hypothesis (H_1)

$$H_1: p > 0.95$$

The alternative hypothesis states that the new algorithm has improved the detection rate.

This is a right-tailed test.

f. Select the Significance Level

$$\alpha = 0.05$$

g. Calculate the Test Statistic

The test statistic for a population proportion is:

$$Z = (\hat{p} - p_0) / \sqrt{[p_0(1 - p_0) / n]}$$

Substituting the values:

$$Z = (0.97 - 0.95) / \sqrt{[0.95(1 - 0.95) / 200]}$$

$$Z = 0.02 / \sqrt{(0.0475 / 200)}$$

$$Z = 0.02 / 0.0154$$

$$Z \approx 1.30$$

Therefore,

$$Z = 1.30$$

h. Calculate the p-value

Since this is a right-tailed test:

$$\text{p-value} = P(Z > 1.30)$$

From the standard normal distribution table:

$$\text{p-value} \approx 0.097$$

i. Compare the p-value with α

$$\text{p-value} = 0.097$$

$$\alpha = 0.05$$

Since:

$$0.097 > 0.05$$

The p-value is greater than the significance level.

Therefore, we fail to reject the null hypothesis.

j. Industry-Oriented Conclusion

The new cybersecurity algorithm achieved a detection rate of 97%, compared to the historical detection rate of 95%. However, the p-value of 0.097 is greater than the significance level of 0.05.

Therefore, there is not enough statistical evidence at the 5% significance level to conclude that the new detection algorithm has significantly improved the detection rate. Although the observed detection rate is higher, additional testing with a larger sample may be required to confirm a significant improvement.

Final Conclusion: Fail to reject H_0 .

2. A hospital introduces a new appointment scheduling system. Before implementation, the average patient waiting time was 42 minutes. After implementation, a sample of 64 patients has an average waiting time of 38 minutes with standard deviation 12 minutes. Test whether the new system significantly reduces waiting time using the p-value approach.

Answer:

a. Identify the Population and Sample

Population: All patients who use the new appointment scheduling system at the hospital.

Sample: 64 patients selected after the implementation of the new appointment scheduling system.

b. Identify the Parameter of Interest

The parameter of interest is the true mean waiting time of patients after implementing the new appointment scheduling system.

Let:

μ = True mean patient waiting time after implementation.

c. Calculate the Point Estimate

The point estimate for the population mean is the sample mean.

$\bar{x} = 38$ minutes

Therefore, the estimated average patient waiting time is:

$\bar{x} = 38$ minutes

d. State the Null Hypothesis (H_0)

$H_0: \mu = 42$ minutes

The null hypothesis states that the new appointment scheduling system has not reduced the average patient waiting time.

e. State the Alternative Hypothesis (H_1)

$H_1: \mu < 42$ minutes

The alternative hypothesis states that the new appointment scheduling system has reduced the average patient waiting time.

This is a left-tailed test.

f. Select the Significance Level

$\alpha = 0.05$

g. Calculate the Test Statistic

Since the population standard deviation is not given, we use a one-sample t-test.

The test statistic is:

$$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$$

Substituting the values:

$$t = (38 - 42) / (12 / \sqrt{64})$$

$$t = -4 / (12 / 8)$$

$$t = -4 / 1.5$$

$$t = -2.67$$

Therefore:

$$t = -2.67$$

Degrees of freedom:

$$df = n - 1$$

$$df = 64 - 1$$

$$df = 63$$

h. Calculate the p-value

For:

$$t = -2.67$$

$$df = 63$$

Since this is a left-tailed test, the p-value is approximately:

$$p\text{-value} \approx 0.005$$

i. Compare the p-value with α

$$p\text{-value} = 0.005$$

$$\alpha = 0.05$$

Since:

$$0.005 < 0.05$$

The p-value is less than the significance level.

Therefore, we reject the null hypothesis.

j. Industry-Oriented Conclusion

The average patient waiting time decreased from 42 minutes to 38 minutes after implementing the new appointment scheduling system. The calculated p-value of approximately 0.005 is less than the significance level of 0.05.

Therefore, there is sufficient statistical evidence to conclude that the new appointment scheduling system significantly reduces patient waiting time.

The hospital can consider the new system effective in improving appointment management and reducing patient waiting time.

Final Conclusion: Reject H_0 . The new appointment scheduling system significantly reduces the average patient waiting time.

3. A social media platform claims that the average daily usage time of its users is 3 hours. A random sample of 100 users has an average usage time of 3.2 hours with standard deviation 0.8 hours. Determine whether the data provide sufficient evidence to reject the company's claim at $\alpha = 0.05$.

Answer:

a. Identify the Population and Sample

Population: All users of the social media platform.

Sample: 100 randomly selected users of the social media platform.

b. Identify the Parameter of Interest

The parameter of interest is the true mean daily usage time of all users of the social media platform.

Let:

μ = True mean daily usage time of all users.

c. Calculate the Point Estimate

The point estimate for the population mean is the sample mean.

$$\bar{x} = 3.2 \text{ hours}$$

Therefore, the estimated average daily usage time is:

$$\bar{x} = 3.2 \text{ hours}$$

d. State the Null Hypothesis (H_0)

$$H_0: \mu = 3 \text{ hours}$$

The null hypothesis states that the average daily usage time of the users is equal to 3 hours, as claimed by the company.

e. State the Alternative Hypothesis (H_1)

$$H_1: \mu \neq 3 \text{ hours}$$

The alternative hypothesis states that the average daily usage time of the users is different from 3 hours.

This is a two-tailed test.

f. Select the Significance Level

$$\alpha = 0.05$$

g. Calculate the Test Statistic

Since the sample size is large, we use the test statistic:

$$Z = (\bar{x} - \mu_0) / (s / \sqrt{n})$$

Substituting the values:

$$Z = (3.2 - 3) / (0.8 / \sqrt{100})$$

$$Z = 0.2 / (0.8 / 10)$$

$$Z = 0.2 / 0.08$$

$$Z = 2.50$$

Therefore:

$$Z = 2.50$$

h. Calculate the p-value

Since this is a two-tailed test:

$$\text{p-value} = 2P(Z > 2.50)$$

From the standard normal distribution table:

$$\text{p-value} \approx 0.0124$$

i. Compare the p-value with α

$$\text{p-value} = 0.0124$$

$$\alpha = 0.05$$

Since:

$$0.0124 < 0.05$$

The p-value is less than the significance level.

Therefore, we reject the null hypothesis.

j. Industry-Oriented Conclusion

The sample of 100 users has an average daily usage time of 3.2 hours, which is different from the company's claimed average of 3 hours. The calculated p-value of approximately 0.0124 is less than the significance level of 0.05.

Therefore, there is sufficient statistical evidence to reject the company's claim that the average daily usage time of its users is 3 hours.

The social media platform may need to update its reported average usage statistics based on further analysis of user activity data.

Final Conclusion: Reject H_0 . There is sufficient evidence to conclude that the average daily usage time is significantly different from 3 hours.

4. A data science team develops a new classification model. The existing model has an accuracy of 82%, while the new model achieves 86% accuracy on a randomly selected test sample of 500 observations. Formulate suitable null and alternative hypotheses and determine whether the improvement is statistically significant using a p-value.

Answer:

a. Identify the Population and Sample

Population: All observations or cases on which the new classification model may be used.

Sample: 500 randomly selected test observations used to evaluate the performance of the new classification model.

The new model achieved an accuracy of 86% on the sample.

b. Identify the Parameter of Interest

The parameter of interest is the true classification accuracy of the new model.

Let:

p = True accuracy of the new classification model.

c. Calculate the Point Estimate

The point estimate for the population proportion is the sample accuracy.

$$\hat{p} = 86\%$$

$$\hat{p} = 0.86$$

Therefore, the estimated accuracy of the new classification model is:

$$\hat{p} = 0.86 \text{ or } 86\%$$

d. State the Null Hypothesis (H_0)

The existing model has an accuracy of 82%.

$$H_0: p = 0.82$$

The null hypothesis states that the new model has not significantly improved the classification accuracy.

e. State the Alternative Hypothesis (H_1)

$$H_1: p > 0.82$$

The alternative hypothesis states that the new classification model has significantly improved the classification accuracy.

This is a right-tailed test.

f. Select the Significance Level

$$\alpha = 0.05$$

g. Calculate the Test Statistic

The test statistic for a population proportion is:

$$Z = (\hat{p} - p_0) / \sqrt{[p_0(1 - p_0) / n]}$$

Substituting the values:

$$Z = (0.86 - 0.82) / \sqrt{[0.82(1 - 0.82) / 500]}$$

$$Z = 0.04 / \sqrt{[0.82 \times 0.18 / 500]}$$

$$Z = 0.04 / \sqrt{(0.1476 / 500)}$$

$$Z = 0.04 / \sqrt{0.0002952}$$

$$Z = 0.04 / 0.01718$$

$$Z \approx 2.33$$

Therefore:

$$Z = 2.33$$

h. Calculate the p-value

Since this is a right-tailed test:

$$p\text{-value} = P(Z > 2.33)$$

From the standard normal distribution table:

$$p\text{-value} \approx 0.0099$$

i. Compare the p-value with α

$$p\text{-value} = 0.0099$$

$$\alpha = 0.05$$

Since:

$$0.0099 < 0.05$$

The p-value is less than the significance level.

Therefore, we reject the null hypothesis.

j. Industry-Oriented Conclusion

The new classification model achieved an accuracy of 86%, compared with the existing model accuracy of 82%. The calculated p-value of approximately 0.0099 is less than the significance level of 0.05.

Therefore, there is sufficient statistical evidence to conclude that the new classification model has significantly improved the classification accuracy.

The data science team can consider deploying the new model because its improvement in accuracy is statistically significant compared with the existing model.

Final Conclusion: Reject H_0 . The new classification model shows a statistically significant improvement in accuracy.

5. A manufacturer claims that its production process has a defect rate of only 2%. From 2,000 randomly selected products, 55 are defective. Perform a hypothesis test at the 5% significance level using both (a) confidence interval and (b) p-value approaches. Compare the conclusions.

Answer:

a. Identify the Population and Sample

Population: All products manufactured using the production process.

Sample: 2,000 randomly selected products.

Out of these 2,000 products, 55 products were found to be defective.

b. Identify the Parameter of Interest

The parameter of interest is the true proportion of defective products produced by the manufacturer.

Let:

p = True defect rate of the production process.

c. Calculate the Point Estimate

The sample proportion is:

$$\hat{p} = x / n$$

$$\hat{p} = 55 / 2000$$

$$\hat{p} = 0.0275$$

Therefore, the point estimate of the defect rate is:

$$\hat{p} = 2.75\%$$

d. State the Null Hypothesis (H_0)

The manufacturer claims that the defect rate is 2%.

$$H_0: p = 0.02$$

The null hypothesis states that the true defect rate is equal to 2%.

e. State the Alternative Hypothesis (H_1)

$$H_1: p \neq 0.02$$

The alternative hypothesis states that the true defect rate is different from 2%.

This is a two-tailed test.

f. Select the Significance Level

$$\alpha = 0.05$$

Therefore, the confidence level is:

$$\text{Confidence Level} = 95\%$$

g. Calculate the Confidence Interval

The formula for the confidence interval of a population proportion is:

$$\hat{p} \pm Z_{\alpha/2} \sqrt{[\hat{p}(1 - \hat{p}) / n]}$$

For a 95% confidence level:

$$Z_{\alpha/2} = 1.96$$

Substituting the values:

$$0.0275 \pm 1.96 \sqrt{[0.0275(1 - 0.0275) / 2000]}$$

$$0.0275 \pm 1.96 \sqrt{(0.02674 / 2000)}$$

$$0.0275 \pm 1.96 \times 0.00366$$

$$0.0275 \pm 0.00717$$

Therefore:

$$\text{Lower Limit} = 0.02033$$

$$\text{Upper Limit} = 0.03467$$

Thus, the 95% confidence interval is:

$$0.0203 < p < 0.0347$$

Or, in percentage form:

$$2.03\% < p < 3.47\%$$

Since 2% is not included in the confidence interval, we reject the manufacturer's claim at the 5% significance level.

h. Calculate the Test Statistic

The test statistic for a population proportion is:

$$Z = (\hat{p} - p_0) / \sqrt{[p_0(1 - p_0) / n]}$$

Substituting the values:

$$Z = (0.0275 - 0.02) / \sqrt{[0.02(1 - 0.02) / 2000]}$$

$$Z = 0.0075 / \sqrt{(0.0196 / 2000)}$$

$$Z = 0.0075 / \sqrt{0.0000098}$$

$$Z = 0.0075 / 0.00313$$

$$Z \approx 2.40$$

Therefore:

$$Z = 2.40$$

i. Calculate the p-value

Since this is a two-tailed test:

$$\text{p-value} = 2P(Z > 2.40)$$

From the standard normal distribution table:

$$\text{p-value} \approx 0.0164$$

j. Compare the p-value with α

$$\text{p-value} = 0.0164$$

$$\alpha = 0.05$$

Since:

$$0.0164 < 0.05$$

The p-value is less than the significance level.

Therefore, we reject the null hypothesis.

k. Compare the Confidence Interval and p-value Conclusions

Using the confidence interval approach, the 95% confidence interval for the defect rate is 2.03% to 3.47%. Since the claimed defect rate of 2% is not included in this interval, we reject the null hypothesis.

Using the p-value approach, the p-value of 0.0164 is less than $\alpha = 0.05$. Therefore, we reject the null hypothesis.

Both the confidence interval and p-value approaches give the same conclusion.

I. Industry-Oriented Conclusion

The observed defect rate in the sample is 2.75%, which is higher than the manufacturer's claimed defect rate of 2%. Both the confidence interval and p-value approaches show that this difference is statistically significant at the 5% significance level.

Therefore, there is sufficient statistical evidence to conclude that the actual defect rate is significantly different from 2%. Since the observed defect rate is higher, the manufacturer should review and improve its production and quality control processes to reduce product defects.

Final Conclusion: Reject H_0 . Both the confidence interval and p-value approaches indicate that the actual defect rate is significantly different from the claimed 2% defect rate.

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	

Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	